



Solved example problem:

(Frequency response analysis)

Let's say we are given the below transfer function:

$$G(s) = \frac{20 \cdot K (s+4)}{s(s+1)(s+7)}$$

K is variable proportional gain.
for $K=1$, let's first find the exact magnitude and phase expressions.

$$\text{With } K=1, G(s) = \frac{20 (s+4)}{s(s+1)(s+7)}$$

put $s = j\omega$ to obtain frequency response.

$$G(j\omega) = \frac{20(j\omega + 4)}{(j\omega)(j\omega + 1)(j\omega + 7)}$$

$G(j\omega)$ is a complex number.
So, it has a magnitude and phase.

$$z = a + jb, \quad |z| = \sqrt{a^2 + b^2}$$
$$\angle z = \theta = \text{phase} = \tan^{-1} \frac{b}{a}$$

$$\text{Magnitude: } |G(j\omega)| = \frac{20 \sqrt{16 + \omega^2}}{\sqrt{\omega^2} \sqrt{1 + \omega^2} \sqrt{49 + \omega^2}}$$

$$|G(j\omega)| = \frac{20 \sqrt{16 + \omega^2}}{\omega \sqrt{1 + \omega^2} \sqrt{49 + \omega^2}}$$

$$\text{Phase : } \angle G(j\omega) = \tan^{-1} \frac{\omega}{4} - \tan^{-1} \frac{\omega}{0} \\ - \tan^{-1} \frac{\omega}{1} - \tan^{-1} \frac{\omega}{7}$$

$$\angle G(j\omega) = \tan^{-1} \frac{\omega}{4} - \tan^{-1} \infty - \tan^{-1} \omega \\ - \tan^{-1} \frac{\omega}{7}$$

(Note: Numerator angles are positive and denominator angles are negative.)

for example, $z_1 = r_1 e^{j\theta_1}$ (polar form)

$$z_2 = r_2 e^{j\theta_2}$$

$$z_3 = r_3 e^{j\theta_3}$$

$$z_4 = r_4 e^{j\theta_4}$$

$$G(j\omega) = \frac{z_1 z_2}{z_3 z_4} = \frac{r_1 e^{j\theta_1} r_2 e^{j\theta_2}}{r_3 e^{j\theta_3} r_4 e^{j\theta_4}}$$

$$= \frac{r_1 r_2}{r_3 r_4} \cdot e^{j(\theta_1 + \theta_2 - \theta_3 - \theta_4)}$$

$$|G(j\omega)| = \left| \frac{r_1 r_2}{r_3 r_4} \right|$$

$$\angle G(j\omega) = \theta_1 + \theta_2 - \theta_3 - \theta_4$$

This explains why the denominator phase is negative.)

Moving forward, let's find the gain cross-over frequency ω_{gc} .

$$|G(j\omega)| = 1$$

$$\Rightarrow \frac{20 \cdot \sqrt{16 + \omega^2}}{\omega \sqrt{1 + \omega^2} \sqrt{\omega^2 + 49}} = 1$$

$$\Rightarrow \frac{400(16 + \omega^2)}{\omega^2(1 + \omega^2)(49 + \omega^2)} = 1$$

$$\Rightarrow 400(16 + \omega^2) = \omega^2(1 + \omega^2)(49 + \omega^2)$$

$$\Rightarrow 6400 + 400\omega^2 = 49\omega^2 + \omega^4 + 49\omega^4 + \omega^6$$

$$\Rightarrow \omega^6 + 50\omega^4 - 351\omega^2 - 6400 = 0$$

This is a sixth order polynomial equation. It has six roots.

Use matlab function "roots" to find the solution. Among six roots, only one root will be positive and real. Others will be negative or imaginary. Positive real root is the solution.

$$\text{Gain cross-over freq.} = \omega_{gc} = 3.64 \text{ rad/s}$$

Now, let's calculate the phase corresponding to the gain cross-over freq.

$$\omega_{gc} = 3.64 \text{ rad/s}$$

$$\angle G(j\omega) = \tan^{-1} \frac{3.64}{4} - 90^\circ - \tan^{-1} 3.64$$

$$- \tan^{-1} \frac{3.64}{7}$$

$$= 42.3^\circ - 90^\circ - 74.63^\circ - 27.47^\circ$$

$$\angle G(j\omega) = -149.8^\circ$$

In order to find the phase margin we need to check how far the above angle from -180° .

$$\text{phase margin} = \text{PM} = -149.8 - (-180^\circ)$$

$$= 30.2^\circ$$

Here, we have positive phase margin. So, the system is stable.

In order to find the phase cross-over frequency ω_{pc} ,

$$\angle G(j\omega) = -180^\circ$$

$$\Rightarrow \tan^{-1} \frac{\omega}{4} - 90^\circ - \tan^{-1} \omega - \tan^{-1} \frac{\omega}{7} = -180^\circ$$

$$\text{as } \omega \rightarrow \infty, \angle G(j\omega) \rightarrow -180^\circ$$

But it never crosses the -180° angle. So, there is no finite phase cross-over frequency.

In this kind of situation, the gain margin is said to be

infinite.

$$\text{Gain Margin} = GM = \infty$$

In general, the following three steps can be used to find the gain margin:

1. Find the phase crossover freq.
2. Evaluate the open loop magnitude at that freq ω_{pc} .

$$3. \text{ Gain margin} = \frac{1}{|G(j\omega_{pc})|}$$

If there is no ω_{pc} , then

$|G(j\omega_{pc})|$ is not defined.

And, Gain Margin = ∞ , at that situation.